What Is UML?

Drawing software before you build it

How do you draw a design?

UML

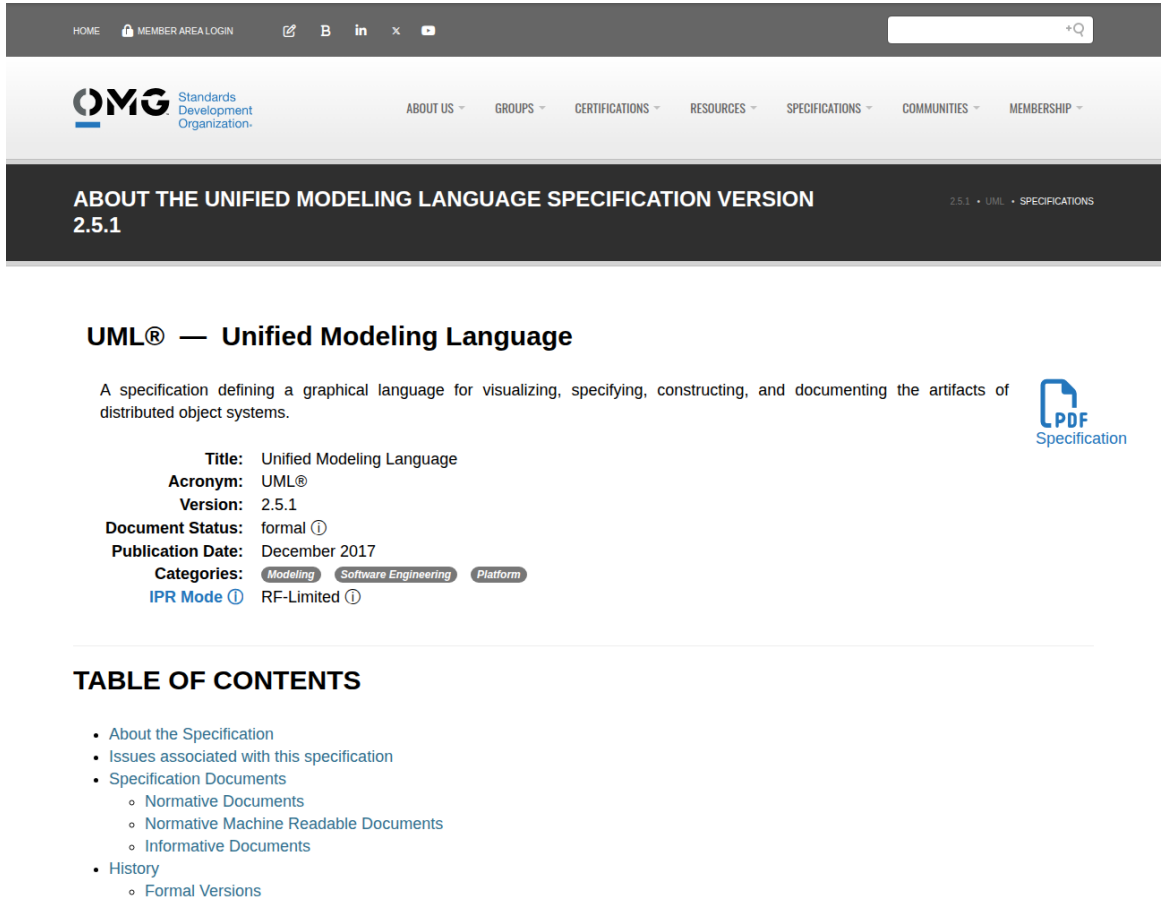
A standard visual language for software designs

The Official Standard

Maintained by the OMG

Current version 2.5.1

Free to read as a PDF



Source

Fourteen Diagrams, Two Families

Structure: what it is	Behaviour: what it does
Class	Use case
Component	Sequence
Deployment	State machine

Class diagram

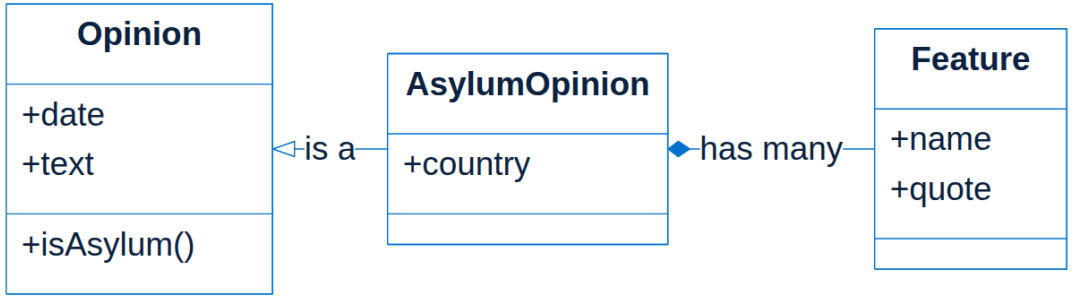
The things in a system and how they relate

A Class Diagram, Concretely

Each box is a class

Each line is a link

Modelling ninth_circuit



Source

Reading a Class Box

Part of the box	What it holds
Top	The class name
Middle	Its attributes
Bottom	Its operations

The Lines Between Classes

Line end	Means
Hollow triangle	Is a kind of
Filled diamond	Owns, and dies with
Hollow diamond	Has, but shares
Plain line	Knows about

Sequence diagram

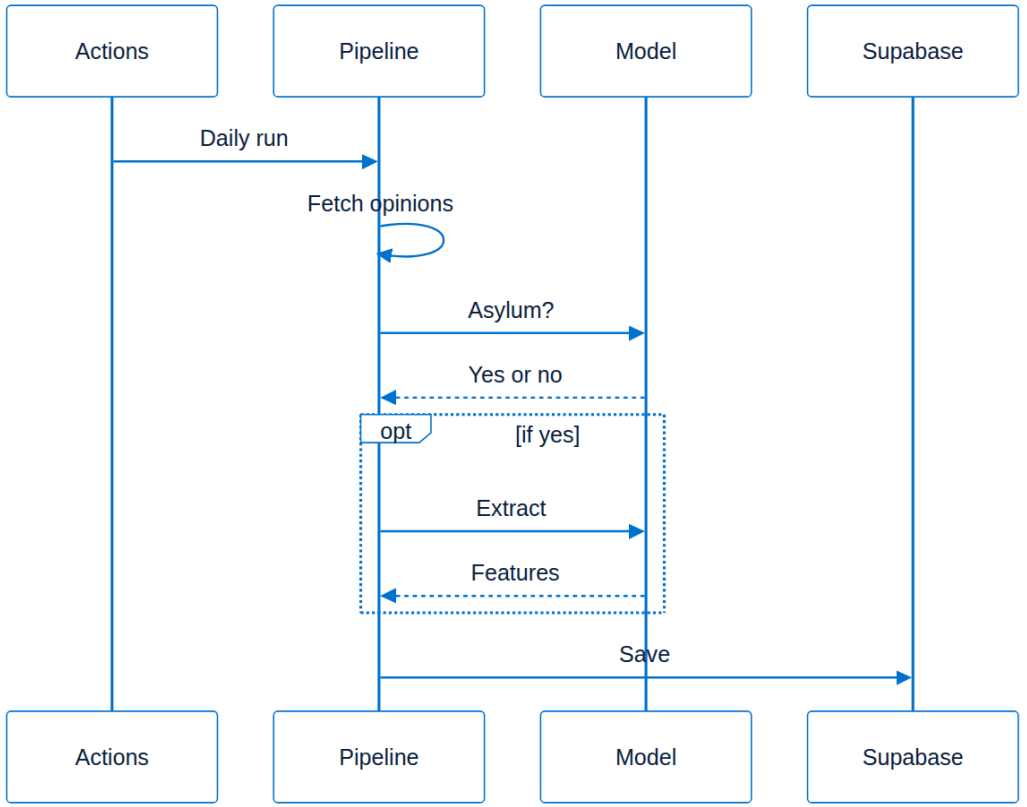
Who talks to whom, and in what order

A Sequence Diagram, Concretely

Time runs downward

Solid arrows: requests

Dashed arrows: replies



Source

Boxes That Change the Flow

opt: happens only if true

alt: one branch or the other

loop: repeats while true

par: happens at the same time

State machine

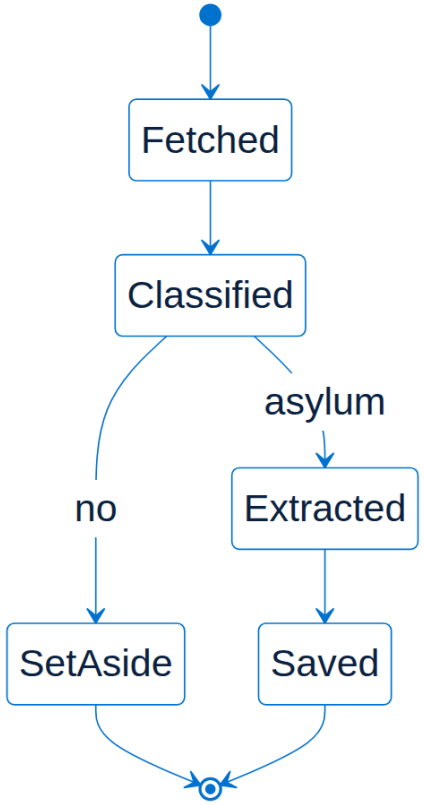
The stages one thing moves through

A State Diagram, Concretely

Filled dot: the start

Ringed dot: the end

Labels say why it moves



Source

Which Diagram, When

If you are asking	Draw a
What is it made of?	Class diagram
Who talks to whom?	Sequence diagram
What stages can it be in?	State diagram
What can a user do?	Use case diagram

Tools for Drawing It

Pen and paper, honestly

Mermaid: diagrams written as text

PlantUML: also text

draw.io: drag and drop

Sources

Slide	Taken from
Specification	omg.org/spec/UML
The diagram types	Wikipedia, UML
Example diagrams	Drawn with Mermaid
What they model	ninth_circuit README

Good Habits

Draw to explain, not to archive

One question per diagram

Leave out what does not matter

Keep it next to the code